

Distance Meter with LCD

Arduino Uno R3 + Tinkercad | HC-SR04 Ultrasonic Sensor + 16x2 LCD Display

Yeh project HC-SR04 ultrasonic sensor use karke distance measure karta hai aur result seedha 16x2 LCD screen par live show karta hai. Code is PDF mein copy-paste ke liye maujood hai, aur sath mein alag Distance_Meter_LCD.ino file bhi di gayi hai jo seedha Tinkercad ya Arduino IDE mein open ki ja sakti hai.

Update: Is version mein LCD ki wiring aapke asal 16x2 LCD module ke physical pinout (GND, VCC, V0, RS, RW, E, DB0-DB7, LED, LED) ke mutabiq correct ki gayi hai. Pehle diagram mein generic "EN", "D4-D7" aur "Backlight A/K" labels use hue the — ab yeh module ke silkscreen labels (E, DB4-DB7, LED/LED) se match karte hain, taake Tinkercad ya real hardware par confusion na ho.

1. Components Used

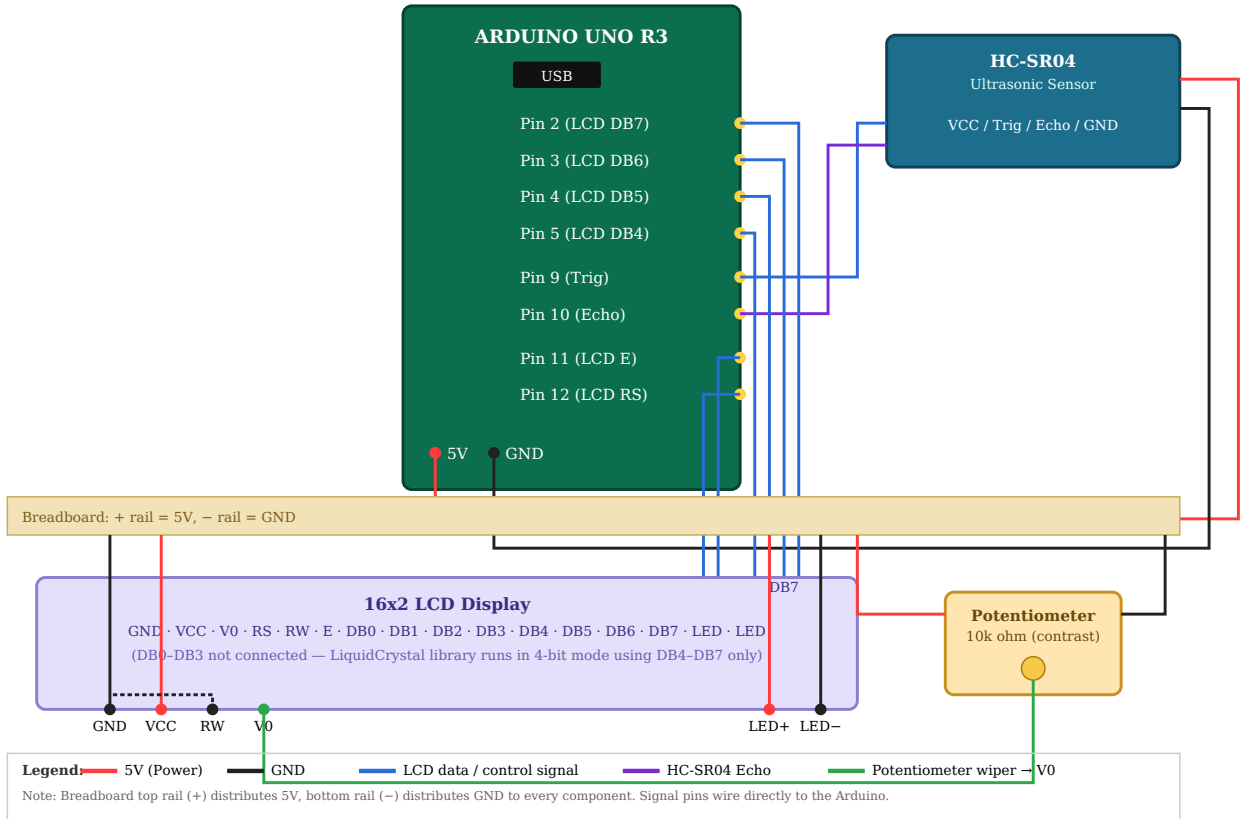
Component	Quantity	Purpose
Arduino Uno R3	1	Main microcontroller board
HC-SR04 Ultrasonic Sensor	1	Measures distance using sound waves
16x2 LCD Display	1	Shows live distance reading
Potentiometer (10k ohm)	1	Adjusts LCD contrast
Breadboard	1	For connections
Jumper Wires	As needed	For wiring all components

2. Circuit Diagram corrected pin labels

White background par color-coded wiring niche di gayi hai (legend diagram ke neeche hai). LCD pin names ab module ke actual silkscreen (GND, VCC, V0, RS, RW, E, DB0-DB7, LED, LED) se match karte hain:

Distance Meter with LCD — Wiring Diagram (Breadboard Layout)

LCD pin labels corrected to match physical module: GND · VCC · V0 · RS · RW · E · DB0-DB7 · LED · LED



3. Build Steps

1. Tinkercad open karo aur naya circuit banao.
2. Arduino Uno, breadboard, HC-SR04, 16x2 LCD, aur potentiometer sab drag karke place karo.
3. HC-SR04 wiring: VCC to 5V, GND to GND, Trig to Pin 9, Echo to Pin 10.
4. LCD wiring (corrected pin names): RS to Pin 12, E to Pin 11, DB4 to Pin 5, DB5 to Pin 4, DB6 to Pin 3, DB7 to Pin 2, GND to GND, VCC to 5V, RW to GND, LED (+) to 5V, LED (–) to GND. DB0–DB3 ko chhod do — inhe kuch bhi connect nahi karna, kyunki LiquidCrystal library 4-bit mode mein sirf DB4–DB7 use karti hai.
5. Potentiometer wiring: dono ends 5V aur GND se, middle pin (wiper) LCD ke V0 se.
6. Code editor open karo aur LiquidCrystal.h library wala code paste karo.
7. Simulation run karo — LCD screen par live distance dikhna chahiye.

4. Full Arduino Code

Note: LiquidCrystal library call mein pin order Arduino side ke digital pins hai (RS, E, DB4, DB5, DB6, DB7) — yeh code pehle se sahi tha, isme koi tabdeeli nahi ki gayi.

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);  // RS, E, DB4, DB5, DB6, DB7

#define trigPin 9
#define echoPin 10

void setup() {
  lcd.begin(16, 2);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  lcd.print("Distance Meter");
  delay(1500);
  lcd.clear();
}

void loop() {
  long duration, distance;

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;

  lcd.setCursor(0, 0);
  lcd.print("Distance:");
  lcd.setCursor(0, 1);
  lcd.print(distance);
  lcd.print(" cm ");

  delay(500);
}
```

Tip: Is code ko seedha copy-paste kar sakte ho, ya sath diya gaya Distance_Meter_LCD.ino file directly Tinkercad/Arduino IDE mein open kar lo.

5. Wiring Summary Table corrected pin names

Component Pin	Full Name	Arduino Pin	Pin Type	Wire Color
HC-SR04 - VCC	Voltage Common Collector	5V	Power	Red
HC-SR04 - GND	Ground	GND	Ground	Black
HC-SR04 - Trig	Trigger	Pin 9	Digital Output	Blue
HC-SR04 - Echo	Echo	Pin 10	Digital Input	Purple
LCD - GND	Ground	GND	Ground	Black
LCD - VCC	Logic Supply Voltage	5V	Power	Red
LCD - V0	Contrast Voltage	Potentiometer wiper	Contrast Control	Green
LCD - RS	Register Select	Pin 12	Digital Output	Blue

LCD - RW	Read / Write	GND	Ground	Black
LCD - E	Enable	Pin 11	Digital Output	Blue
LCD - DB0	Data Bus Line 0	Not connected	—	—
LCD - DB1	Data Bus Line 1	Not connected	—	—
LCD - DB2	Data Bus Line 2	Not connected	—	—
LCD - DB3	Data Bus Line 3	Not connected	—	—
LCD - DB4	Data Bus Line 4	Pin 5	Digital Output	Blue
LCD - DB5	Data Bus Line 5	Pin 4	Digital Output	Blue
LCD - DB6	Data Bus Line 6	Pin 3	Digital Output	Blue
LCD - DB7	Data Bus Line 7	Pin 2	Digital Output	Blue
LCD - LED (+)	Backlight Anode	5V	Power	Red
LCD - LED (–)	Backlight Cathode	GND	Ground	Black
Potentiometer - Ends	Power Terminals	5V / GND	Power	Red / Black

Note: Arduino Uno ke Pin 0 aur Pin 1 serial communication ke liye reserved hain, isliye is project mein use nahi kiye gaye. DB0–DB3 LCD module par maujood hain lekin 4-bit mode mein unhe connect nahi karna hota — yeh table ab module ke asal 16-pin layout ko sahi tarah reflect karti hai. Wire colors upar circuit diagram ke legend se match karte hain.

6. Kaise Kaam Karta Hai

- HC-SR04 wahi tarike se distance measure karta hai — Trig pulse bhejna, Echo pulse ka duration parhna, aur formula se cm mein convert karna.
- Result Serial Monitor ki bajaye LiquidCrystal.h library ke zariye 16x2 LCD screen par likha jata hai.
- lcd.begin(16, 2) LCD ko initialize karta hai (16 columns, 2 rows). lcd.setCursor(col, row) batata hai ke agla text kahan likhna hai, aur lcd.print() asal value LCD par display karta hai.
- Potentiometer LCD ke V0 pin se juda hota hai taake screen ka contrast manually adjust kiya ja sake — agar text nazar na aaye to pot ko ghuma kar theek kiya jata hai.
- RW pin ko direct GND se joda gaya hai kyunki hum LCD ko hamesha 'write mode' mein rakhte hain (kabhi LCD se data parhna nahi hota).
- DB0–DB3 pins ko chhod diya gaya hai — LiquidCrystal library 4-bit mode mein sirf DB4–DB7 use karti hai, is liye baaki 4 data lines ko connect karne ki zaroorat nahi.
- Har 500 milliseconds mein loop() naya distance measure karta hai aur LCD ki dusri row update karta rehta hai, isliye reading real-time mehsoos hoti hai.

7. Potentiometer Wiring — Detail Mein Samajhiye

Potentiometer ke 3 pins hote hain aur har pin ka alag kaam hota hai:

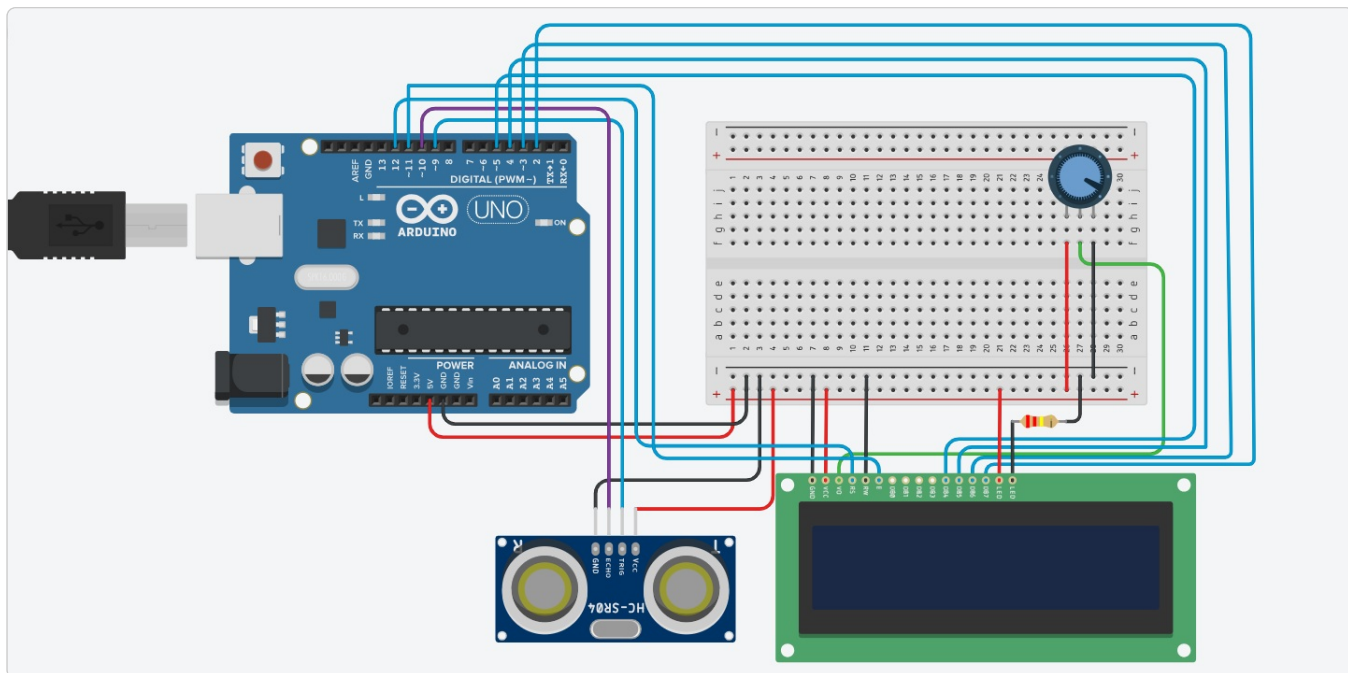
Pin	Connection	Kaam
End 1	5V	Resistive track ke ek sirre ko power deta hai
End 2	GND	Resistive track ke dusre sirre ko ground deta hai
Middle (Wiper)	LCD V0	Track se adjustable voltage nikal kar LCD ko deta hai

- Potentiometer ek variable resistor hai — andar ek resistive track hoti hai jiske do sirre (ends) hain.
- Jab ek end ko 5V aur dusre end ko GND se jorte hain, to us poori track par 0V se 5V tak voltage gradually distribute ho jata hai (ek taraf 5V, dusri taraf 0V).
- **Middle pin (wiper)** ek chhota sa slider hota hai jo knob ghumane par us track par move karta hai. Yeh wiper jahan bhi ho, wahan ka voltage nikal deta hai — kabhi 0V ke qareeb, kabhi 5V ke qareeb, ya beech mein koi bhi value (0–5V ke darmiyan).
- Yeh wiper wala output LCD ke **V0 (contrast)** pin mein jata hai — isi se contrast control hota hai.

Practical matlab: Knob ghumane se V0 pin ko milne wala voltage change hota hai — isi se LCD ka contrast kam ya zyada hota hai. Agar screen par text bilkul nazar na aaye (blank ya bohot dark dikhe), to potentiometer ghuma kar adjust karte hain jab tak text clearly dikhne lage. Simple tareeke se: **don't ends power supply karte hain, middle pin us power ka ek "adjustable hissa" nikal kar LCD ko deta hai.**

8. Tinkercad Breadboard View

Neeche actual Tinkercad simulation ka breadboard layout hai, jisme corrected LCD wiring aur backlight resistor (220Ω) bhi dikhaya gaya hai:



Arduino + Tinkercad Project Series — Distance Meter with LCD (LCD pinout corrected to match physical module)